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Multi-criteria based stable clustering technique for vehicular ad-hoc networks

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Vehicular Ad-hoc Network (VANET) plays an important role in the realization of the Intelligent Transportation System (ITS) which leads to a plethora of new applications such as traffic/drivers safety, traffic management, and infotainment. However, the high mobility and dense distribution of the vehicles on the road cause problems for routes management and network scalability. These issues can be effectively resolved by employing clustering techniques which not only supports scalability of network thereby eradicating broadcast storms caused by packets flooding but also simplifies medium access control. Despite its numerous advantages, cluster formation and sustainability of clusters in high mobility scenarios is one of the major challenges in VANETs. To improve stability of the cluster, efficient cluster head selection in accordance with the various traits of VANETs among candidate vehicles is desired. In this paper, a Multi Criteria based Stable Clustering technique (MCSC) for VANETs is proposed. The proposed MCSC technique includes a novel feature of average Past Cluster head Lifetime (PCL) along with relative speed, distance and degree of connectivity for efficient selection of Cluster Head (CH) and elects Substitute Cluster Head (SCH) to maintain the stability of cluster structures. Apart from CH selection, cluster formation, maintenance and clusters merging features are also incorporated. A simulation based performance evaluation is conducted to evaluate efficacy of the proposed technique through different parameters and benchmarking with state-of-the-art. Demonstrated by simulation results, the proposed MCSC technique outperforms existing clustering techniques in VANETs with respect to key stability indicators. Specifically, MCSC improves the cluster-head and cluster-member lifetimes by approximately 16% and 14.5%, respectively, compared to existing approaches. Furthermore, the proposed technique reduces the re-clustering rate by about 13%, indicating enhanced cluster stability under dynamic vehicular conditions.

Keywords ITS, PCL, Stable clustering, VANET

Vehicular Ad-Hoc Network (VANET), a sub-class of Mobile Ad-Hoc Network (MANET), is a self-organized wireless network of vehicles along the road. With the advancements in automobile technology, VANET lays the foundation for the implementation of ITS application. These applications include traffic management, traffic safety and sharing infotainment messages[1]. In VANETs, vehicles are equipped with on-board units (OBUs) which allows them to communicate with each other, called V2V communication. Likewise, vehicles are also able to communicate with deployed infrastructure e.g. road side unit (RSU), thereby connecting to central traffic management. Figure 1 shows communication architecture of VANET.

Compared to MANETs, vehicles in VANETs are highly mobile and have variable density which causes frequent changes in network's topology[2]. Due to the dynamic nature of the network, VANETs pose several challenges, including scalability to support system and application-level services, as well as security and privacy[3,4]. Researchers have scrutinized the idea of clustering in VANETs which aims to transform flat network into hierarchical network by forming groups of vehicles called clusters via some sort of clustering techniques. Each cluster has number of cluster members (CMs) and a leader, known as cluster head (CH). The CH, acts as an access-point thereby providing a kind of virtual infrastructure, and is responsible for inter and intra cluster(s) communication. In this way, benefits of infrastructure-based network are availed without the cost of infrastructure deployment[5]. A clustering structure within the network reduces the communication overhead, supports routing[6], and security[7]. However, it is inevitable that the highly dynamic topology of VANETs will

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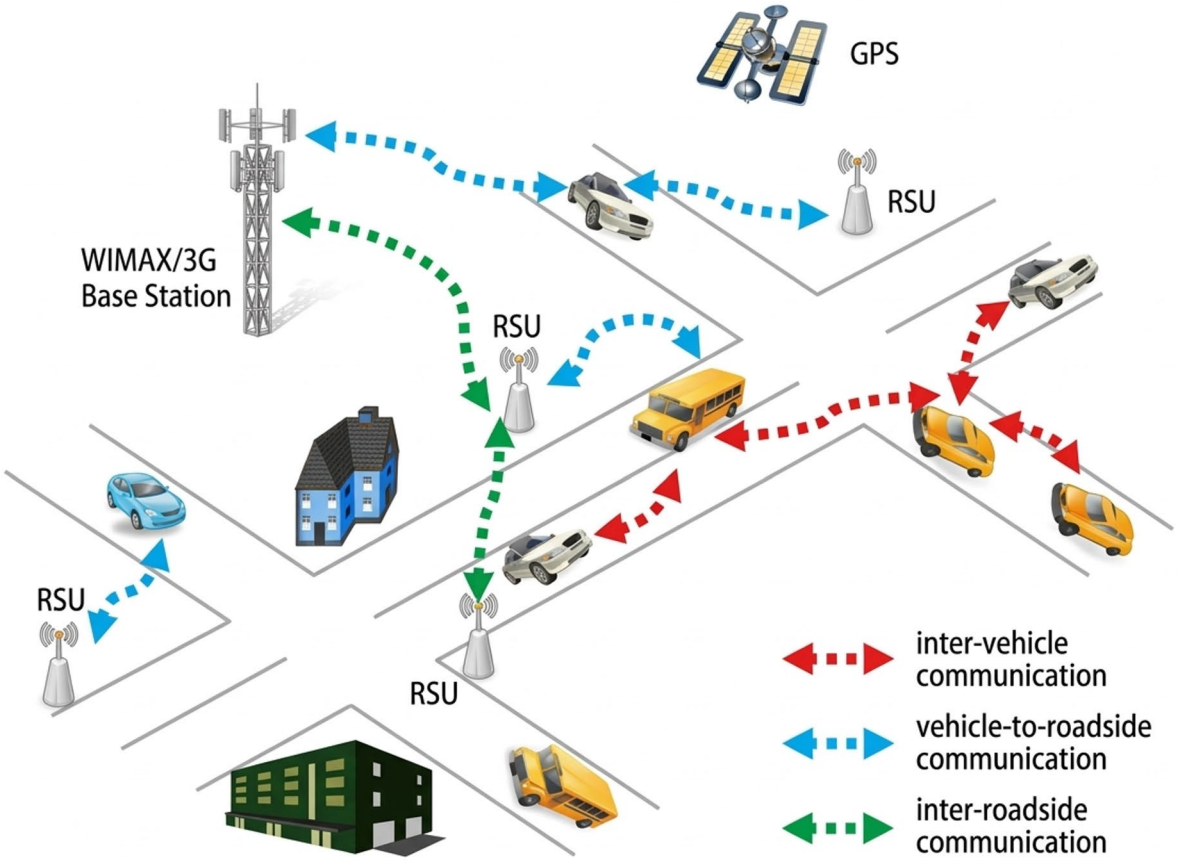


Fig. 1. Communication architecture of VANET[9].

interrupt the already established clusters and result in increased communication overhead for clusters maintenance[8]. One of the important desired attributes of clustering techniques especially in VANETs is to avoid frequent re-clustering thereby reducing the management overhead.

Clustering in VANETs provides a stable architectural framework that facilitates subsequent routing, MAC, and trust management operations. Through hierarchical communication via stable cluster heads, clustering not only reduces control overhead but also enables more reliable and long-lasting routes even under high mobility scenarios. Likewise, clustering greatly assists MAC operations by coordinating channel access and avoiding collisions. In the security domain, clustering offers a framework for localized authentication, misbehavior detection, and trust evaluation. More broadly, clustering enables effective cross-layer optimization in VANETs by jointly enhancing routing stability, improving MAC-layer efficiency, and supporting robust trust management. However, the effectiveness of all these operations fundamentally depends on the stability of the formed clusters. The primary focus of this research is to achieve stable clustering in VANETs without explicitly addressing routing, MAC, or trust management mechanisms, as improved cluster stability inherently supports and enhances these downstream operations.

In this paper, we propose a stable clustering technique for VANETs to construct stable clusters such that to maximize lifetime of both cluster-heads and cluster-members while minimize re-clustering rate. In order to attain stability of clusters, firstly we propose a Connectivity LifeSpan (CLS-based) neighbors filtering scheme to filter out unstable neighboring vehicles. Secondly, an efficient CH selection among stable neighbors is conducted on the basis of relative mobility, degree of connectivity and past cluster life. Thirdly, the selection of substitute CH is incorporated to avoid frequent clustering in case of temporary disconnection of CMs with respective CH due to speed variation. These strategies enable the proposed clustering technique to attain relatively more stability than other state-of-the-art. It is worthwhile to mention that the high-level architecture of our proposed MCSC technique shares functional blocks with existing paradigms, the essential novelty lies in the integrated design, and uniquely unified by stability history i.e. PCL. Unlike existing schemes combining instantaneous metrics, the PCL serves as a historical filter, pre-qualifying inherently stable vehicles before weighted metrics are applied thereby fundamentally altering the CH election outcomes. Furthermore, the backup cluster head is an integral part of our design, working in continuity with the PCL-based primary cluster head to enhance stability and provide focused resilience against transient disconnections. Similarly, clusters merging is conditioned

on stability rather than proximity alone, avoiding instability caused by conventional approaches. Thus, while functional blocks appear classic, the design philosophy, component interdependence, and stability-driven metrics collectively distinguish our work as an integrated stability framework.

The rest of this paper is structured as follows: "Related work" describes previous work related to cluster formation and maintenance in VANETs. "Materials and methods" presents our proposed technique. "Performance evaluation" includes performance analysis of the proposed technique by taking into account clustering performance metrics and comparison with recent techniques. The "Limitations and future work" section highlights the current constraints of the study and outlines potential directions for further research to address these limitations. Finally, the paper is concluded in "Conclusion" section.

Related Work

Clustering techniques—partitioning nodes into CHs and CMs on the basis of certain criteria are being in use for quite some time. They are particularly applicable in MANETs and dozens of clustering techniques have been proposed by various researchers. However, due to high mobility of vehicles and frequent topological changes, clustering techniques proposed for MANETs are not directly applicable in VANETs. For instance, Fan et al.[10,11] attempted to apply MANET techniques in VANETs to form clusters and to improve broadcasting performance in inter vehicle communication, but their approaches could not adapt to traffic dynamics.

Despite great diversity in the literature, clustering techniques nearly follow the same procedural flow. The fundamental clustering procedure includes cluster head selection and cluster maintenance schemes. In VANETs, due to high dynamic nature of vehicles, it is difficult for vehicles to communicate with each other in the form of cluster for extended times and maintain the topology. In this scenario, re-clustering and high cluster maintenance cost will reduce efficiency of network and invalidate the purpose of clustering. Therefore, to form stable clusters and sustain their stability, are the vital issues in clustering techniques for VANETs. Researchers have proposed many clustering techniques based on mobility metrics to form stable clusters. This section presents a critical review of relevant clustering techniques for VANETs.

Some studies aim to achieve stable clustering along with reliable and efficient communication. In Ref.[12], proposed a clustering mechanism based on iterative Trimmed and Truncated Mean (iTTM) algorithm and Criteria Importance Through Inter Criteria Correlation (CRITIC) technique aiming to improve the clustering stability. It attains the stated aim in two steps: In the first step, iTTM is used to collect the temporal relationships among the vehicles. In the second step, the CH is selected based on multi-decision CRITIC using consensus trust score, connectivity level, link lifetime and relative speed. It struggles to maintain stable clusters under high mobility because frequent changes in vehicle positions and speeds trigger constant re-elections, reducing cluster stability. Another technique aiming to achieve the goal of the reliable and stable CH selection is proposed in Ref.[13]. The proposed framework is based on Adaptive Neighboring Behavior-Incentivization (ANB-I), where the CH selection process is based on the Multi-Criteria Multi-Decision (MCMD) using link lifetime, neighborhood degree and relative average speed. Trust-based Stable Clustering (TSC)[14] is the mechanism that also aims to select stable CH. The CH selection process is based on multiple parameters including degree, trust value, position, speed, and remaining travel time of the vehicles. These schemes slightly improve the stable clustering but introduces more overheads due to weighing multi-criteria and trust computation.

Some researchers have used the parameter of link lifetime (LLT) to sample surrounding neighbors and form clusters with stable neighbors. MDMAC[15] introduced LET (Link Expiration Time) in messages by the vehicles, to find k-hop neighbors and avoid high-speed CH candidates. The main idea is to reduce number of cluster changes and avoid re-clustering to achieve cluster stability. Density based clustering introduced in Ref.[16] further extends the LET mechanism thereby adding link quality measurement through SNR to assess the suitability of CH to form stable clusters. Ji et al.[17] aims to improve efficiency and reliability by organizing vehicles into clusters and designating cluster heads for data forwarding. While the approach enhances communication performance under moderate mobility, it relies on relatively static CH selection and assumes predictable link durations. In highly dynamic VANET scenarios, where vehicle speeds and positions change rapidly, the already formed clusters quickly become outdated, resulting in frequent re-clustering.

In most of the techniques, there is one CH for each cluster. But, in scenarios where due to speed variation or change in direction, the CH remains unavailable, re-clustering is inevitable. In order to maintain the stability of network and avoid re-clustering, many researchers have proposed different techniques to select secondary cluster head that immediately switches its role as CH once the predecessor leaves the cluster, so that the cluster does not break. A stability-based clustering technique (SBCA)[18] aims to decrease the communication overhead for the cluster formation and maintenance and to increase cluster lifetime. While SBCA selects CH based on link stability to reduce re-clustering, it relies heavily on predicted link lifetimes and relative vehicle speeds. In highly dynamic traffic scenarios, where vehicles frequently change speed, direction, or lane, such predictions lead to incorrect decisions thereby causing frequent re-clustering and instability. Additionally, SBCA does not fully account for rapid topology changes in dense or heterogeneous traffic, limiting its effectiveness in real-world high-mobility conditions.

In Ref.[19] proposed a clustering technique which considers the behavior of vehicles for efficient cluster head selection and also employs a backup CH to preserve the stability of cluster structures. But, CH selection is purely being made on basis of speed of vehicles which does not yield stable clusters. DHC[20] is a clustering technique which uses mobility metrics and communication link quality metrics which includes the SNR and LET for cluster formation. Moreover, secondary CH is selected for cluster maintenance and cluster merging feature is also incorporated to prolong the life of cluster. The drawback of this technique is that it results in increased overhead while evaluating cluster merging options because all CMs are taken into confidence before making final decision. Furthermore, in case of connection of CM with CH and SCH, CM simply joins the next available

CH as a member node without assessing whether it may itself qualify to become the CH. Lacking this self-assessment feature contradicts the formation of stable clusters.

Some of the existing techniques in literature make use of threshold and weight based mechanisms for CH selection. TB and UFC[21,22] are some examples of threshold based clustering technique. The vehicle that firstly announces itself as a CH wins the competition and turns out to be CH. The drawback of this type of schemes is that it may potentially ignore the most eligible CH which affects stability of cluster. In weight-based method, each vehicle is assessed by different metrics to act as CH, such that the one with high value is selected as CH. In this regard, authors in Ref.[23] proposed a clustering scheme to form clusters based on relative distance and velocity of the vehicles and future position by using Gauss Markov Model. But, this probabilistic approach is vulnerable to degraded network performance in case of high variation in mobility. In Ref.[24,25] cluster head selection is on account of mobility metrics which include relative velocity, distance and acceleration. But, there is Infrastructure based cluster maintenance and reforming which contradicts with the typical nature of ad-hoc network and increases maintenance overhead.

Some researchers have added reputation and trust as metrics along with mobility metrics for cluster head selection. TACR[26] takes into account velocity and position along with trust metrics to calculate cluster head selection metric. Trust metrics ensure capability of cluster head and also provide secure communication within cluster. VWCA[27] selects a cluster head on the basis of degree of connectivity, vehicle direction, an entropy value evaluated from the mobility of vehicles, and a distrust level. MOFA[28] takes into consideration the direction, position, velocity and degree of vehicles, ID of lane and reputation of each node for stable topology of VANETs. But, reputation score is based on number of times vehicle perform the task of CH, this may increase reputation score of vehicle to be selected even for short period of time. In order to deal with this, we introduced PCL to examine the consistency of CH through its duration. Behura et al.[29] proposed a multi-objective cluster-based routing approach aiming to enhance security and routing efficiency in VANETs. While effective in ensuring secure communication, the method does not explicitly address cluster stability under highly dynamic urban traffic conditions. Its evaluation focuses on network performance metrics rather than clusters longevity, making it more suitable for highway or low-mobility scenarios. Kumar et al.[1] introduced Enhanced Clustering based Efficient Relay Vehicle (ECERV) selection framework for VANETs. The CH selection is mainly influenced by instantaneous network conditions, which can result in frequent cluster reconfigurations in high mobility scenarios.

Based on literature review, it is revealed that majority of existing clustering techniques for VANETs lead to unstable clustering especially in high mobility scenarios which increases communication overhead. Taking this into account, we aim to develop a stable clustering technique that is not only in accordance with the traits of VANETs but also avoids frequent re-clustering caused by high mobility of vehicles.

Materials and methods

In this section, we propose a cluster-based vehicle-to-vehicle (V2V) communication technique without any need of deployed infrastructure such as RSUs for communication among vehicles. We assume that each vehicle is equipped with On board Unit (OBU) containing a Global Positioning System (GPS), a Dedicated Short-Range Communication (DSRC) wireless unit and a Processing Unit (PU). The vehicle uses the GPS to obtain its real-time position, velocity and moving direction and PU is used to run the clustering technique on each vehicle separately. Vehicles periodically share their information with each other via hello messages. These messages include vehicle's identity, current velocity, current position, moving direction, and past CH Lifetime (PCL). The vehicle which has not played any CH role in the past, its PCL value is considered as zero. Our technique is applicable for both urban and highway traffic. The roads may have multiple lanes and are bi-directional but with no intersections.

We assume a fully trusted operational environment in this work where vehicles share messages without any malicious intent. The main purpose of this study is to evaluate stability of clusters under realistic mobility conditions and therefore we do not consider adversarial threats such as forgery of messages, Sybil, or spoofing attacks. Nevertheless, we acknowledge that real-world deployment would require integrating lightweight authentication and integrity protection mechanisms to ensure the reliability of the information exchanged during cluster formation and maintenance.

Our proposed model MCSC is a distributed technique. Every vehicle runs this technique independently after getting information of other vehicles through hello messages. In this section, we illustrate the vehicles' state with and without clustering, neighbors' filtration strategy and techniques for cluster formation, maintenance and merging. The general flow chart of our proposed clustering technique is showed in Fig. 2.

Initial cluster formation

Initially upon deployment of vehicles along the roads they are unclustered as no clusters exist at this point. Vehicles are said to be in unclustered state (UC) at this stage. Later on, the vehicles might acquire cluster head (CH), cluster member (CM), substitute cluster head (SCH) and/or bridge node (BN) status, depending upon the roles they are delegated. Figure 3 shows an example of clusters formation in VANETs using the proposed MCSC technique.

A brief description of these states is given as follows:

- UC: Initial state of all vehicles when the vehicle does not belong to any clusters.
- SN: Those neighbors which qualify connectivity lifespan (CLS) criteria to be considered as stable neighbor vehicles and remain connected with each other for a predetermined time.
- CH: The head of the cluster to supervise and coordinate all communications among its members.
- CM: The vehicle which is a one-hop neighbor of a CH.

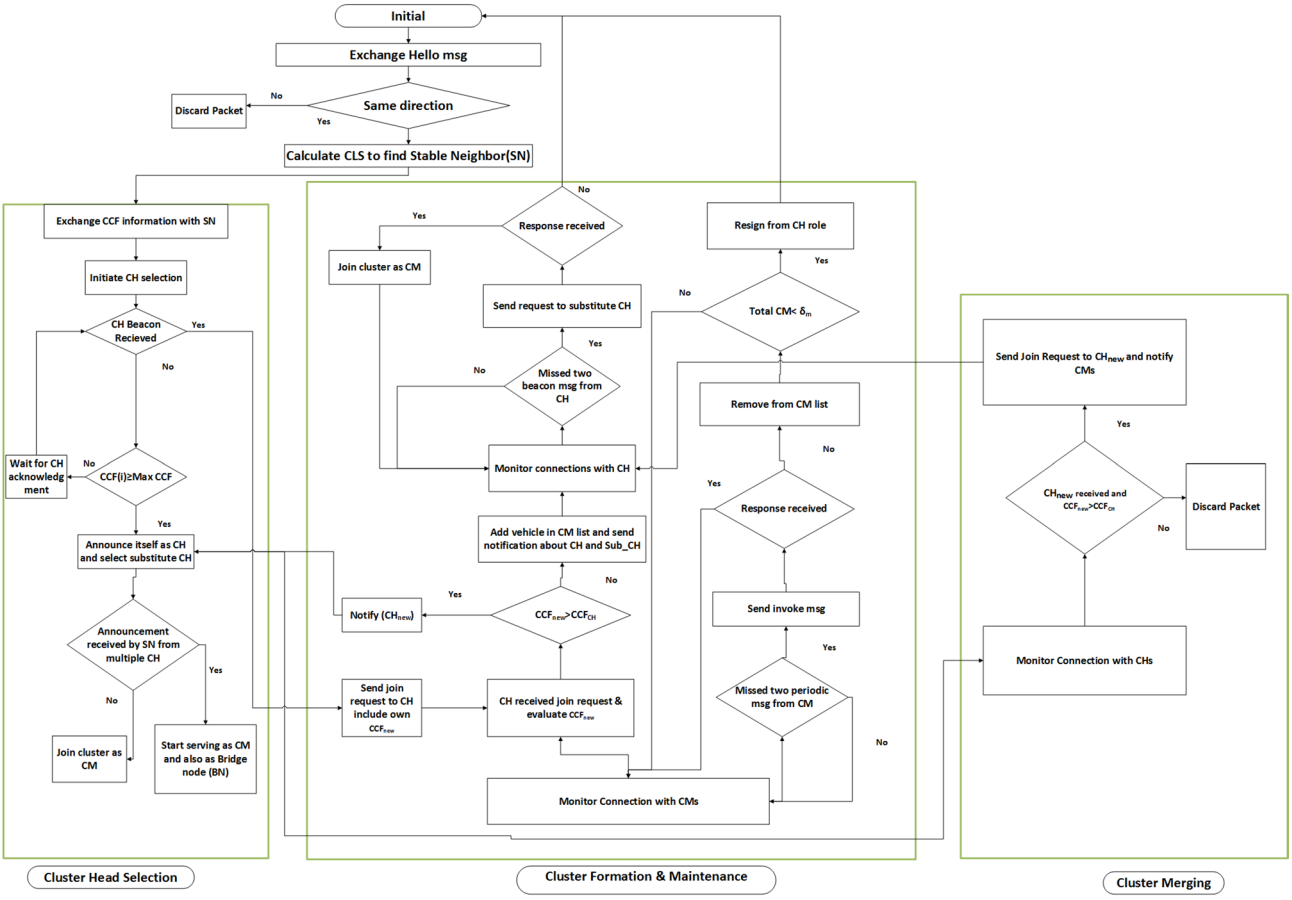


Fig. 2. Flow chart of the proposed MCSC technique.

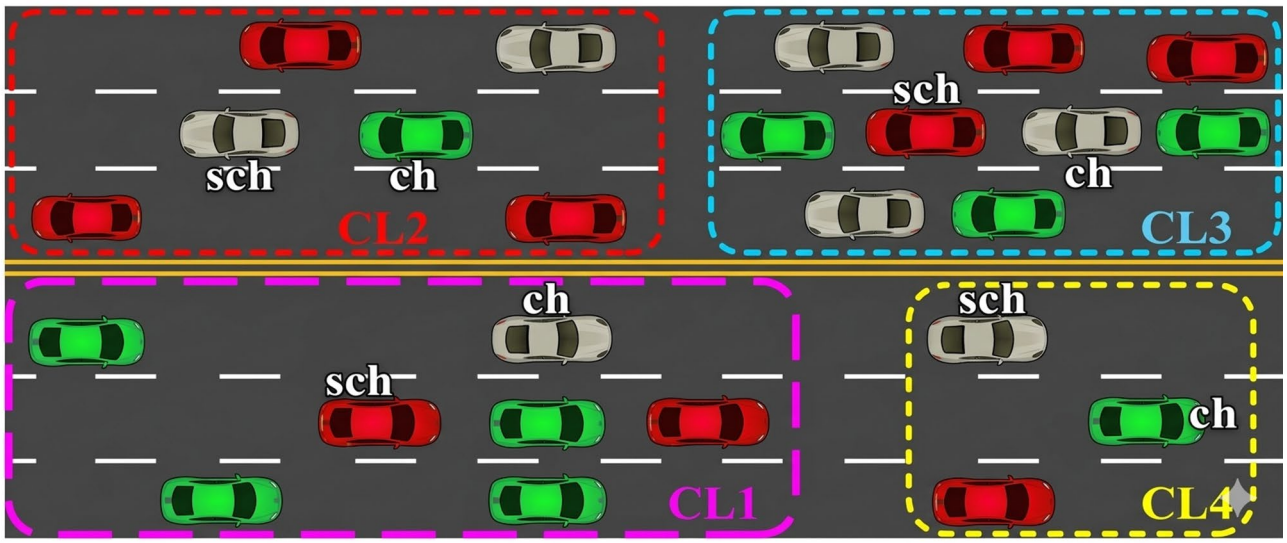


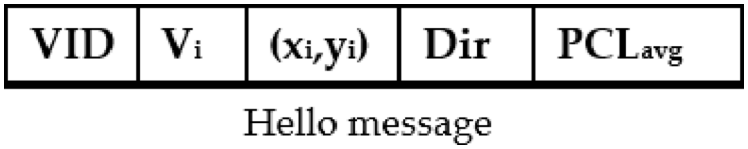
Fig. 3. Clusters Formation in VANET[9].

- SCH: The vehicle which operates as backup of the CH when the CM loses connection with the CH.
- BN: Bridge Node is such a vehicle which is receiving beacon messages from multiple CH and can help in interconnecting or passing messages between adjacent CH.

Vehicles switch their state triggered by different events. The detail of vehicles state switching is presented in the following section. The notations used in this paper are stated in Table 1. Vehicles seeking to be clustered will go through various stages namely neighborhood discovery, CLS based neighbor filtration and cluster head selection. Detailed description of these stages is given in the following sub-sections.

Neighborhood discovery

Initially, every vehicle which joins the road is in the UC state. Any vehicle wishing to become part of VANET, starts sending periodic Hello messages to announce its presence and to share its information with the surrounding neighbors. This information includes the vehicle ID (VID), velocity (Vi), two-dimensional position (x, y), direction (Dir) and average past cluster head lifetime (PCL_{avg}).



Once a Hello message is received from neighboring vehicles, not all of them are supposed to be grouped in the same cluster. Firstly, considering the direction of vehicles, vehicles moving in opposite direction will be removed from the neighbors list. Next CLS based neighbors filtering scheme is employed to classify neighbors into stable and unstable neighbors considering transmission range of the vehicle, relative velocity and distance between the vehicles.

Notation	Description
UC	Unclustered vehicle
CH	Cluster head
SCH	Substitute cluster head
CM	Cluster member
BN	Bridge node
SN	Stable neighbor
VID	ID of vehicle
Dir	Direction of vehicle
Vi	Velocity of vehicle
State(V_i)	State of vehicle
R	Communication range
PCL	Past cluster head lifetime
CLS	Connectivity lifespan
Deg_conn	Degree of connectivity of the vehicle indicating number of connected stable neighbors
CCF	Composite clustering factor
JOIN_REQ	Join request message sent from CM to respective CH
JOIN_RESP	Join response message sent from CH to respective CM
CH election_msg	Message shared among SN to participate in CH election
CH_Beacon	Cluster Head Beacon Message broadcasted periodically
Invoke_msg	Invoke message from Cluster Head to Cluster Member and is used as last resort to verify connectivity
CM_list	List of cluster members maintained by CH
δ_{SN}	Threshold for stable neighbor
δ_m	Threshold for members in Cluster

Table 1. Summary of notations along with their descriptions.

CLS-based neighbor filtration

To improve clusters stability, we introduce CLS-based neighbor filtration scheme to pick out stable neighbor (SN) from neighboring vehicles. Connectivity Lifespan (CLS) represents the predicted time span in which two neighboring vehicles may stay connected. CLS is calculated using Eq. (1). Here, Δdij and Δvij represent the difference of distance and velocity between two vehicles Vi and Vj, respectively. R represents the transmission range of a vehicle. CLS helps to estimate the connection sustainability. Larger CLS value indicates that connection is more sustainable.

$$\mathrm{CLS_{ij}} = \frac{R - \Delta\, d_{ij}}{\Delta\, v_{ij}} \tag{1}$$

This filtration process is presented in Algorithm 1. In the start, each vehicle Vi gathers information of its neighboring vehicles. Then it filters out unstable neighbors and selects stable neighbor (SN) on the basis of CLS value. A set of stable neighbors SNi, containing several entries SNi(j) of vehicles is maintained for each vehicle Vi.

```
Input: UC
Output: SNi;
(1)  for each vehicle Vi where State(Vi) = UC do
(2)      if Vi receives a Beacon message from Vj then
(3)          if Dir(Vi) = Dir (Vj) then
(4)              Vi calculates CLSij;
(5)          end if
(6)      end if
(7)      if CLSij > δSN then
(8)          if Vj ∈ SNi then
(9)              Vi updates SNi(j)
(10)         else
(11)             Vi adds a new SNi(j) to SNi
(12)             Send Deg_msg to SNi
(13)         end if
(14)     end if
(15) end for
```

Algorithm 1. Stable neighbors filtration

The expression for CLS estimation assumes constant relative velocity and linear trajectory over the short prediction interval, a widely adopted approximation in VANET clustering literature that remains reasonably accurate over typical beaconing intervals (100 ms to a few seconds). We acknowledge that real-world maneuvers such as lane changes, sudden acceleration, and turns can affect CLS accuracy. However, our framework incorporates several practical mitigations: (1) directional filtering restricts CLS calculation to same-direction vehicles (Algorithm 1, line 3), eliminating invalid opposite-direction encounters; (2) frequent updates recalculate CLS upon every beacon reception (Algorithm 1, lines 4 and 9), ensuring continuous reflection of current mobility; (3) threshold-based selection (δSN) filters out vehicles with uncertain predictions; and (4) multi-

criteria robustness combines CLS with Degree of Connectivity and Past Cluster Lifetime, reducing reliance on any single metric.

Cluster head selection

After filtering stable neighbors, vehicles automatically start CH selection process. We define an eligibility parameter namely Composite Clustering Function (CCF), to assess the fitness of a vehicle to act as a cluster head. CCF is formulated in Eq. (2). CLS_{ij} denotes connectivity lifespan with stable neighboring vehicles, Degi denotes the number of stable neighbors of a specific vehicle and PCL gives the value of past cluster head lifetime of vehicle which is a valuable parameter to evaluate its eligibility. A higher PCL value indicates that the vehicle has experience in forming stable clusters, which supports more effective cluster-head selection and improves overall network performance. It is worthwhile to note that in homogeneous traffic conditions where connectivity, density and mobility are fairly uniform, all the three metrics contribute equally toward achieving stable cluster head selection and the overall clustering stability, and therefore equal weightages of 0.333 are assigned to the corresponding coefficients α, β, and γ. For heterogeneous or dynamic traffic conditions, vehicles compute their weightages for α, β, and γ locally based on observed CLS values, neighbors density and PCL history to ensure that stability, connectivity, and experience contribute proportionally to the current network conditions. Equations (3), (4), and (5) are employed to compute values of α, β, and γ in realistic traffic scenarios.

$$CCF_i = \alpha . \ \text{avg}CLS_i + \beta .Deg_i + \gamma .avgPCL_i \tag{2}$$

$$\alpha = \frac{\text{avg}CLS_i}{\text{avg}CLS_i + Deg_i + avgPCL_i} \tag{3}$$

$$\beta = \frac{Deg_i}{\text{avg}CLS_i + Deg_i + avgPCL_i} \tag{4}$$

$$\gamma = \frac{avgPCL_i}{\text{avg}CLS_i + Deg_i + avgPCL_i} \tag{5}$$

In Eq. (2), α, β, and γ are normalized weighted coefficients i.e. $\alpha + \beta + \gamma = 1$ and reflect the dominant factor in local traffic conditions. In the CH selection process, the stable neighbors get to know the CCF of their peers by sharing their own via the CH selection message. Vehicle with highest CCF among the neighbors announce itself as CH according to Algorithm 2. Along with that it also updates other vehicles about substitute cluster head (SCH). If multiple vehicles get identical score for CCF, the proposed MCSE technique employs First-Come-First-Serve (FCFS) mechanism to resolve the conflict and finalize the CH selection.

Hello	VID	CCF_i

CH selection message

```
Input: Set of SN
Output: Set of CH,BN
(1)   while Vi initiate CH selection do
(2)         if Vi receives CH _Beacon from CHj then
(3)               goto Cluster Formation & Maintenance
(4)         else
(5)               Vi calculates CCFi
(6)         end if
(7)         if CCFi ≥ MAXVm∈ SN(CCFi) then
(8)               State(Vi)→CH
(9)               Vi broadcasts CH_Beacon    & update SCH
(10)   end if
(11)  if CH_Beacon >1 received by Vi then
(12)  State(Vi)→ BN
(13)  end if
(14) end   while
```

Algorithm 2. Cluster head selection.

Cluster formation and maintenance

Due to high mobility of vehicles in VANET, there is frequent change in state of vehicles, which causes maintenance overhead. So, the aim of cluster formation & maintenance is to maintain the cluster topology as stable as possible in different scenarios. A set of procedures are proposed to sustain cluster topology in high dynamic nature of network. Algorithm 3 governs the process considering these different scenarios.

I. Join request from UC vehicle.

UC receives beacon message from neighboring CH, it sends join request to the CH with its CCF. CH compares its clustering factor with the incoming join request vehicle; if the value is greater than the CH that means the new UC is more capable to manage the cluster, then CH notifies the UC that it can act as CH. The state of UC changes into CH and the previous CH is selected as substitute SCH, otherwise CH adds the new UC into CM list and its state is set as CM.

II. CH lost connection with CM.

After cluster formation, CH and CM monitor connection with each other. In this scenario, when CH fails to receive two consecutive beacon messages from its CM, it sends an invoke message to CM to verify its status. If it receives response from CM then it monitors connection with the member otherwise removes it from the CM list. CH also checks the count of members in CM list, if it is less than threshold δm, then CH resigns from its role and become UC.

```
Input: Set of UV, CM, CH
Output: Set of CM, CH
(1) for each vehicle Vi where State(Vi) = UV do
(2)         if Vi receives CH_ACK or Beacon from CHj then
(3)                 Vi unicasts JOIN_REQ to CHj
(4)                 CHj evaluate CCFi
(5)                 if CCFi> CCFCHj
(6)                         State(Vi)→ CH then
(7)                         goto Cluster head Selection
(8)                 else
(9)                         State(Vi)→ CM   then
(10)                        Monitor connection with CH
(11)                end if
(12)        end if
(13) end for
(14) for    each vehicle Vi where State(Vi) = CM do
(15)        if   Beacon msg received by CM from CHj < 2 then
(16)                CM   unicasts   JOIN_REQ to SCH
(17)                if CM receives response then
(18)                        Join Cluster & Monitor connection with CH
(19)                else
(20)                         State(Vi)→UV
(21)                 end if
(22)         end if
(23)    end for
(24)    for    each vehicle Vi where State(Vi) = CH do
(25)            if periodic message received from CM <2    then
(26)                    Send Invoke_msg to CM
(27)                    if   CH doesn't receive invoke response    then
(28)                            Remove CM from CM_list
(29)                    end if
(30)            end if
(31)    end for
```

Algorithm 3. Cluster formation and maintenance.

While monitoring connection with CH, when CM misses two consecutive beacon messages from CH, it sends a join request to SCH to make sure cluster remain intact even in the absence of selected CH. After getting response from SCH, CM joins the cluster and monitors connection with SCH as CH. While the SCH mechanism effectively reduces re-clustering events, it is worth mentioning that its performance depends on communication reliability. The proposed MCSC incorporates a tolerance mechanism requiring two consecutive missed beacons before triggering SCH takeover which prevents false transitions due to transient link disconnection. However, if a CM's join request to the SCH or the SCH's response is lost during takeover, the CM reverts to UC state, causing SCH takeover to fail and triggering full re-clustering overhead. Additionally, if SCH response delay exceeds the CM's timeout, the node unnecessarily transitions to UC despite SCH availability, highlighting a trade-off between detection speed and delay tolerance.

IV. Clusters merging.

Parameters	Value/range	Remarks
Road length	30 KM	Urban scenario
Traffic direction	Bi-directional	3 Lanes per direction
Simulation duration	200 s	Per experiment
Iterations	10	Results averaged for robustness
Number of vehicles	50–300	Varied across the experiments
Vehicles' speed	10–30 m/s	Varied across the experiments
Wireless transmission range	200–500 m	Varied across the experiments

Table 2. Summary of simulation parameters.

Clusters formed by vehicles moving in different lanes along the road may overlap with each other. These overlapping moving clusters may cause intra cluster communication and management overhead. In order to overcome this problem, cluster merger technique is proposed. When the distance between two CH is less than transmission range R, then it is convenient to merge overlapped clusters. Algorithm 4 explains this process, in which when CH receives beacon message from neighboring CH it compares its CCF value with the neighboring CH. If neighboring CH has greater value of CCF, then CH sends join request to neighboring CH and notifies its cluster member to join the neighboring cluster.

Input: Two Clusters
Output: Merged Cluster
(1) for each vehicle V_i where State(V_i) = CH do
(2) CH_i received beacon msg from CH_j
(3) if $CCF_{(CHj)} > CCF_{(CHi)}$
(4) CHi unicasts JOIN_REQ to CHj & notify CM
(5) State (CH_i) → CM
(6) goto Cluster formation & maintenance
(7) end if
(8) end for

Algorithm 4. Clusters merging.

Computational complexity and scalability analysis

The proposed MCSC is fully distributed and ensures scalability in large-scale deployments. The decision-making process of each vehicle depends entirely on local information from adjacent vehicles, communicated through the periodic exchange of Hello or beacon messages. The computational overhead for each vehicle is estimated as follows: neighboring filtering and CLS computation incurs O(N); Adaptive weight computation requires O(1); and CCF calculation and CH/SCH selection results in O(N) complexity. For being fully distributed in nature, the proposed MCSC technique results in overall computational complexity of O(N) thereby making it independent of the total number of vehicles. Moreover, the communication overhead of the proposed MCSC technique in the form of control messages exchanges like Hello, CH selection, and SCH notifications are limited to local broadcast. This localized and distributed design enables MCSC to maintain stable clustering performance even in large-scale VANETs scenarios.

Performance evaluation

In this section, we describe simulation setup and results. We used Veins simulator for simulation based performance evaluation of the proposed MCSC technique against ECERV[1] and TSC[13]. Both ECERV and TSC aim to form stable clusters for VANETs thereby making use of multiple parameters.

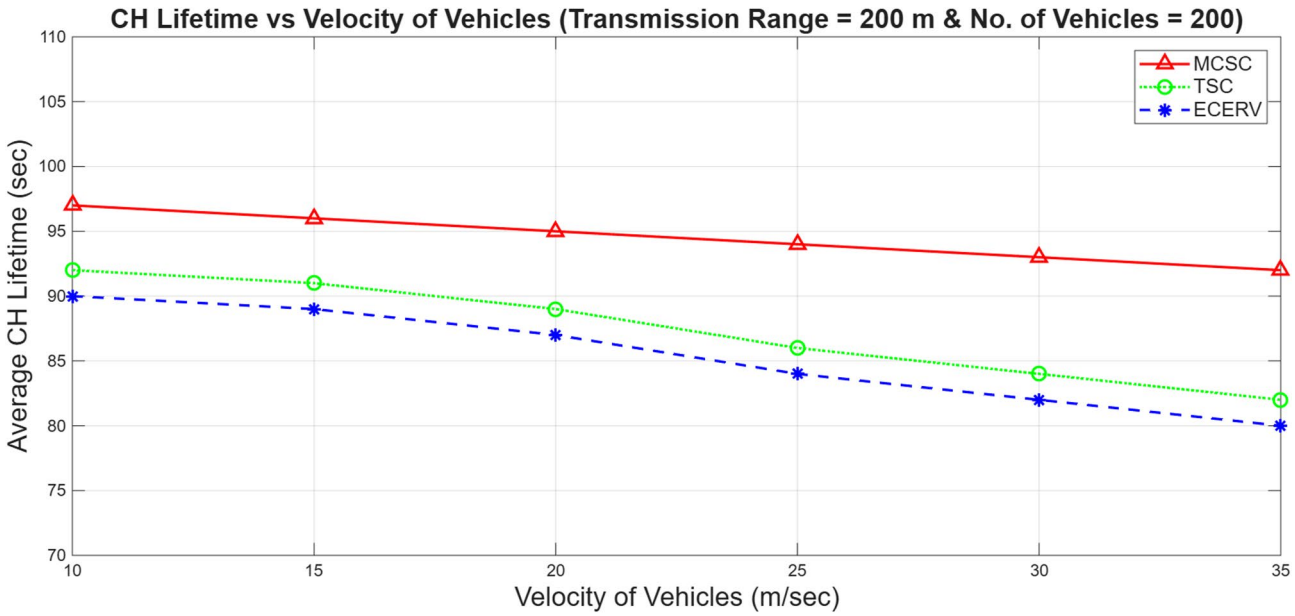


Fig. 4. Impact of velocity of vehicles on CH lifetime.

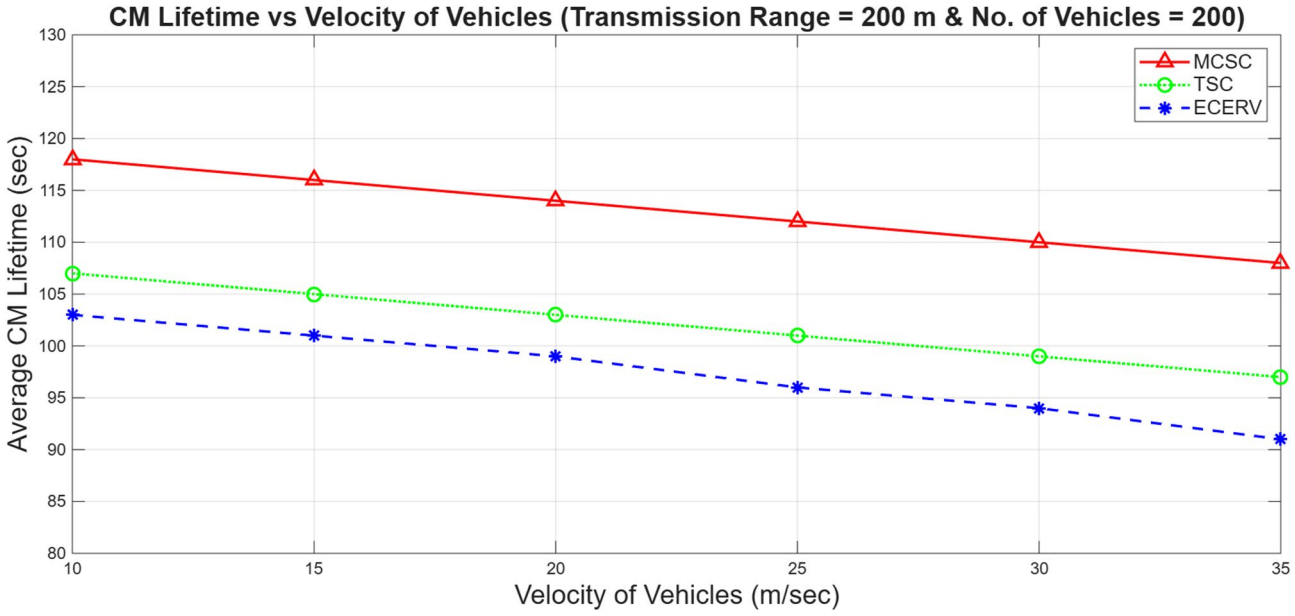


Fig. 5. Impact of velocity of vehicles on CM lifetime.

Simulation Setup

For VANETs simulation in an urban setting, we considered a road length of 30.0 km having bi-directional traffic where each direction is comprised of three lanes. In all our experiments, we ran the simulation for 200 s with different deployments of vehicles. Every experiment was repeated 10 times and the results were averaged out for robustness purposes. Likewise, the total number of vehicles was varied from 50 to 300 having velocity ranging from 10 m/s to 35 m/s. Similarly in different set of experiments, wireless transmission range of vehicles is set in the range from 200 m to 500 m. A summary of the simulation parameters is presented in Table 2.

Performance metrics

Stability index of the proposed MCSC technique is measured and evaluated in terms of average CH duration, average CM duration and re-clustering rate. Furthermore, the performance evaluation in terms of aforementioned metrics is carried out in various network dynamics including the impact of vehicle velocity, the number of vehicles and the vehicles transmission range.

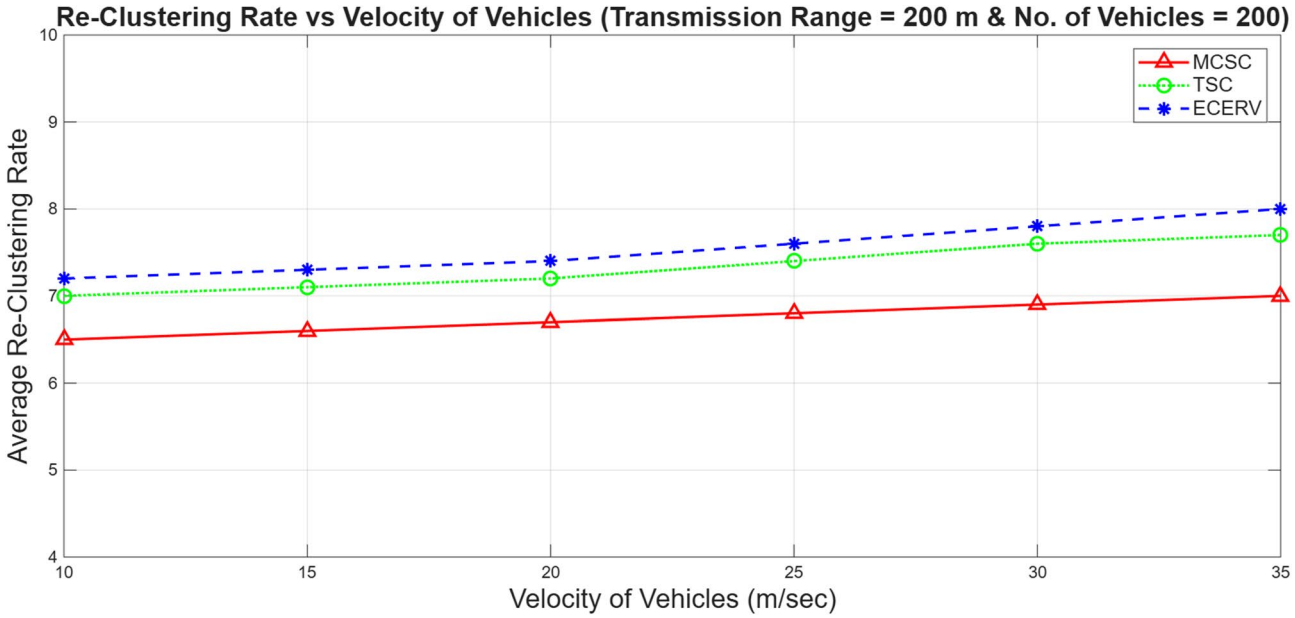


Fig. 6. Impact of velocity of vehicles on re-clustering rate.

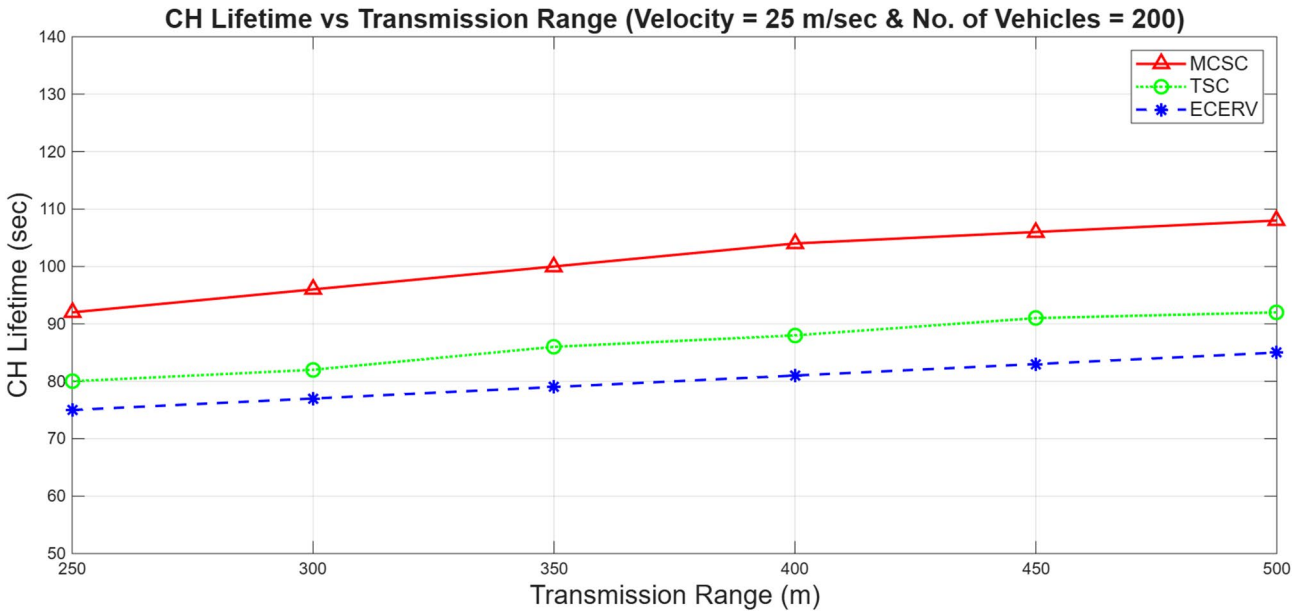


Fig. 7. Impact of transmission range of vehicles on CH lifetime.

For any clustering technique especially in VANETs, variation in velocity of vehicles has a direct impact on the stability of clusters and therefore it should be taken into consideration for evaluation purposes. Likewise, the stability of network by varying the number of vehicles also needs to be assessed. Similarly, the transmission range of vehicles would also have an impact towards stability of links and thus the clusters. A brief description of performance metrics in context of stable clustering is given as follows:

I. Average CH lifetime.

CH lifetime is the duration for which vehicle becomes a CH till the time when it acquires a CM role or leaves the cluster's premises. In this regard, higher average CH lifetime is desired for more stable clusters.

II. Average CM lifetime.

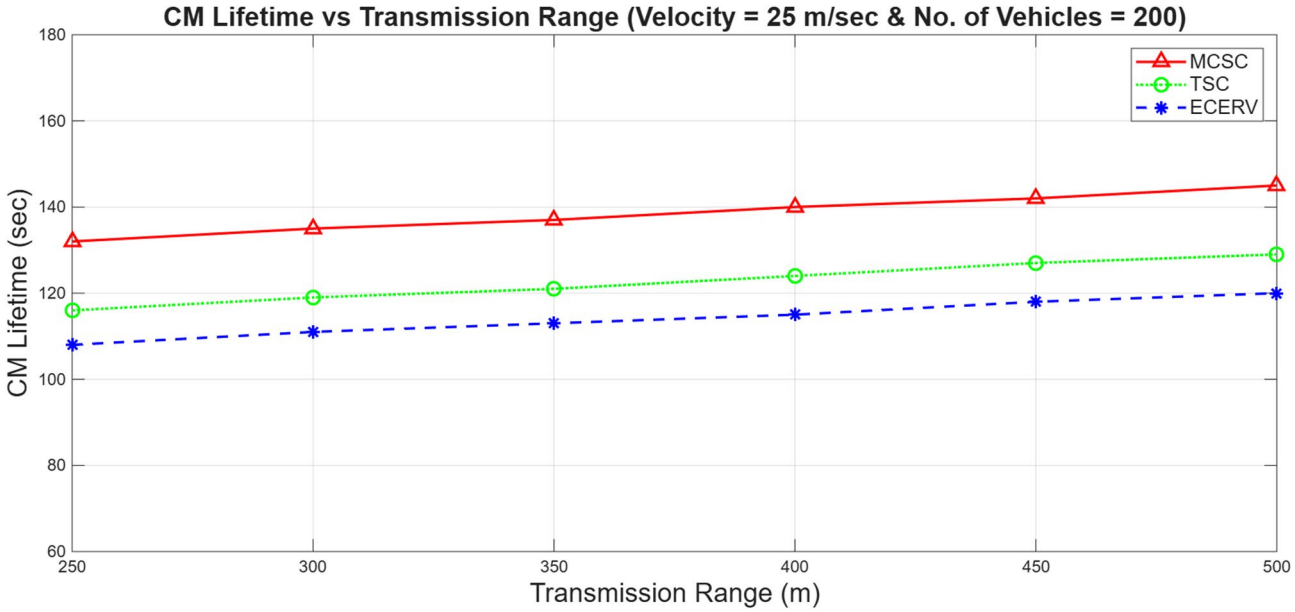


Fig. 8. Impact of transmission range of vehicles on CM lifetime.

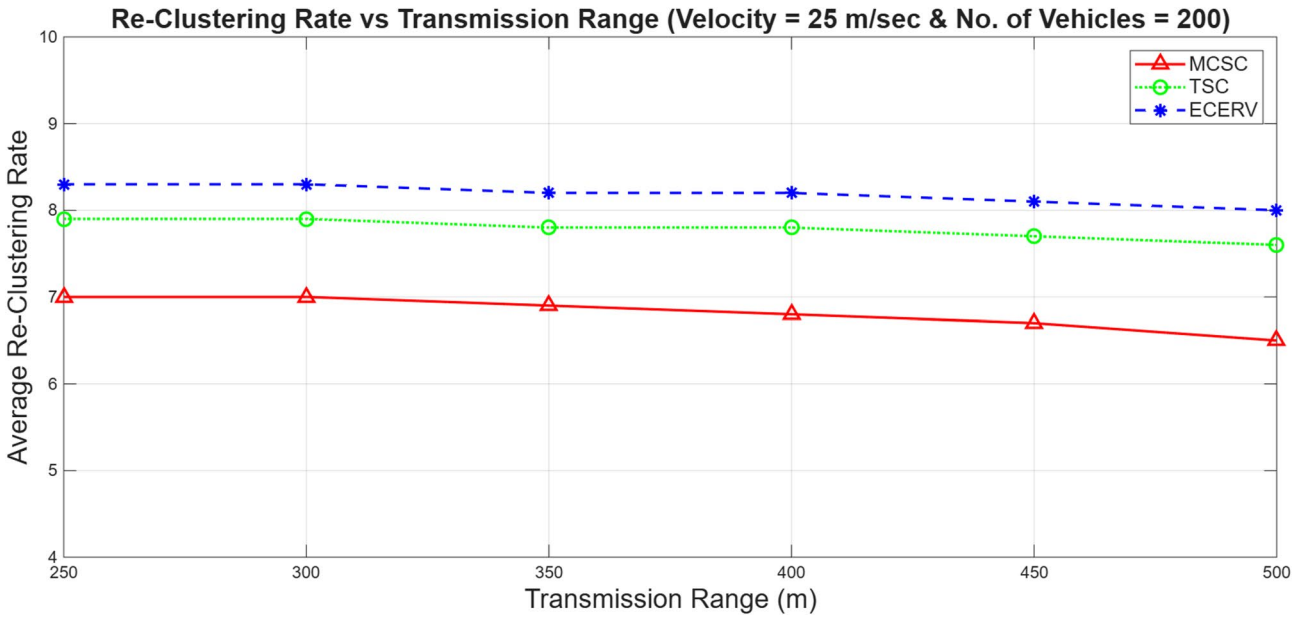


Fig. 9. Impact of transmission range of vehicles on re-clustering rate.

CM lifetime represents the time period for which a CM remains in the same cluster. Likewise, average CM lifetime represents the time duration that every cluster member remains associated with any CH during simulation time. Like average CH lifetime, it also reflects the stability of clusters for vehicles being part of some clusters and are not left unassociated due to frequent re-clustering.

III. Average re-clustering rate.

It indicates how frequently re-clustering is performed during simulation time. This also demonstrates the case when the cluster is completely dissolved and all vehicles acquire the state of UC. Re-clustering events inherently trigger cluster reformation procedures and associated control message exchanges; therefore, a lower re-clustering rate indirectly reflects reduced network overhead. Throughout simulations, improvements in re-clustering rate pertains to indirect reduction of network overhead through enhanced clustering stability, rather than a direct measurement of overhead.

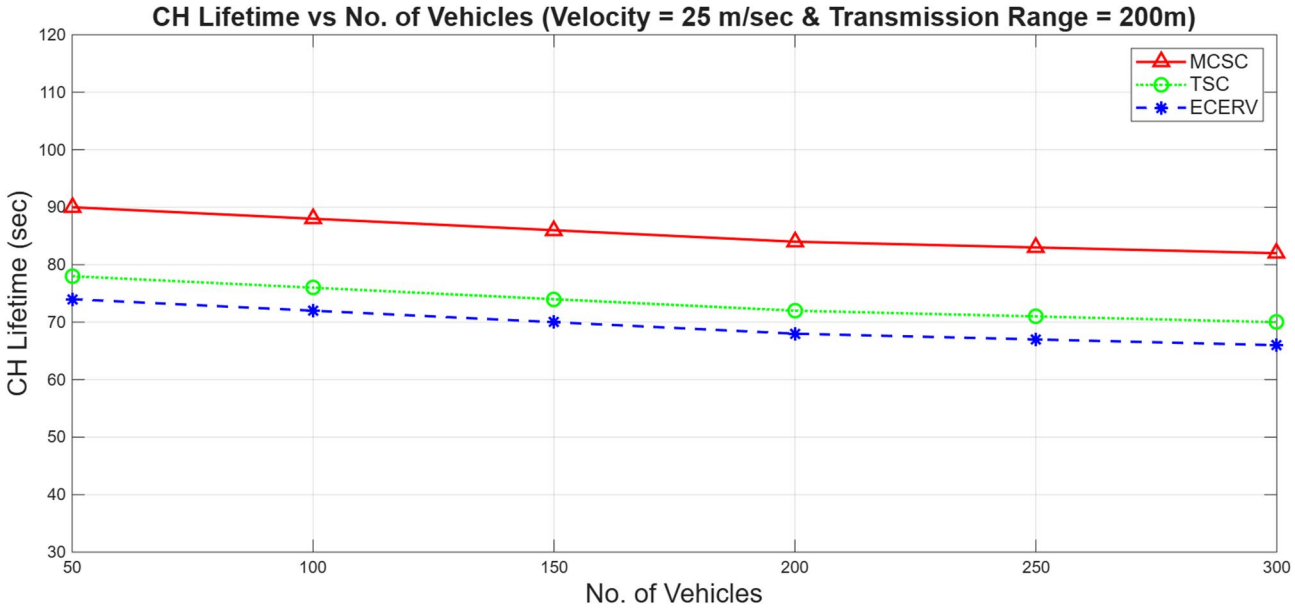


Fig. 10. Impact of number of vehicles on CH lifetime.

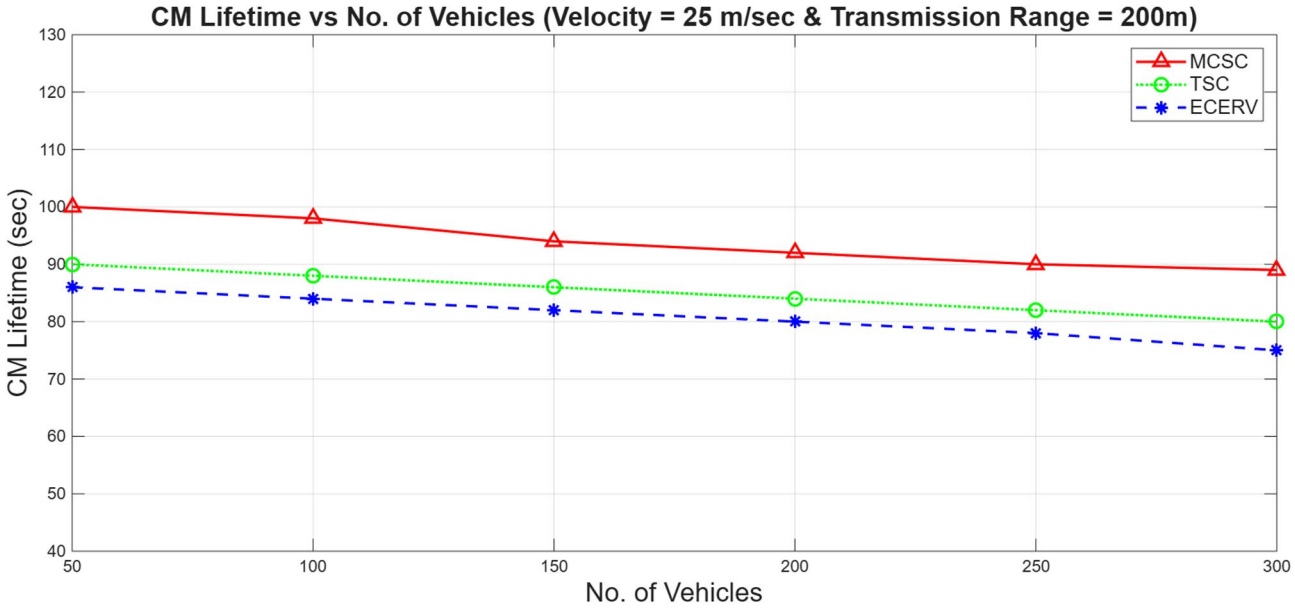


Fig. 11. Impact of number of vehicles on CM lifetime.

Simulation results and discussion

The following subsections describe the performance of MCSC in terms of aforementioned metrics.

I. Impact of velocity of vehicles.

This subsection represents the impact of velocity of vehicles on the performance of MCSC and other techniques in terms of CH duration, CM duration and re-clustering rate. In this set of experiments, we varied velocity of vehicles from 10 to 35 m/s while keeping values of R and N as 200 m and 200 respectively. Figure 4 represents impact of velocity on CH lifetime. In these results we observed that with the increase in number of vehicles, lifetime of CH comparatively decreases in all the three techniques. This is because; increase in velocity causes decrease in CLS which ultimately makes difficult for CH to retain connection with their neighboring vehicles for a long period and network topology become unstable due to its dynamic nature. It is also observed that our

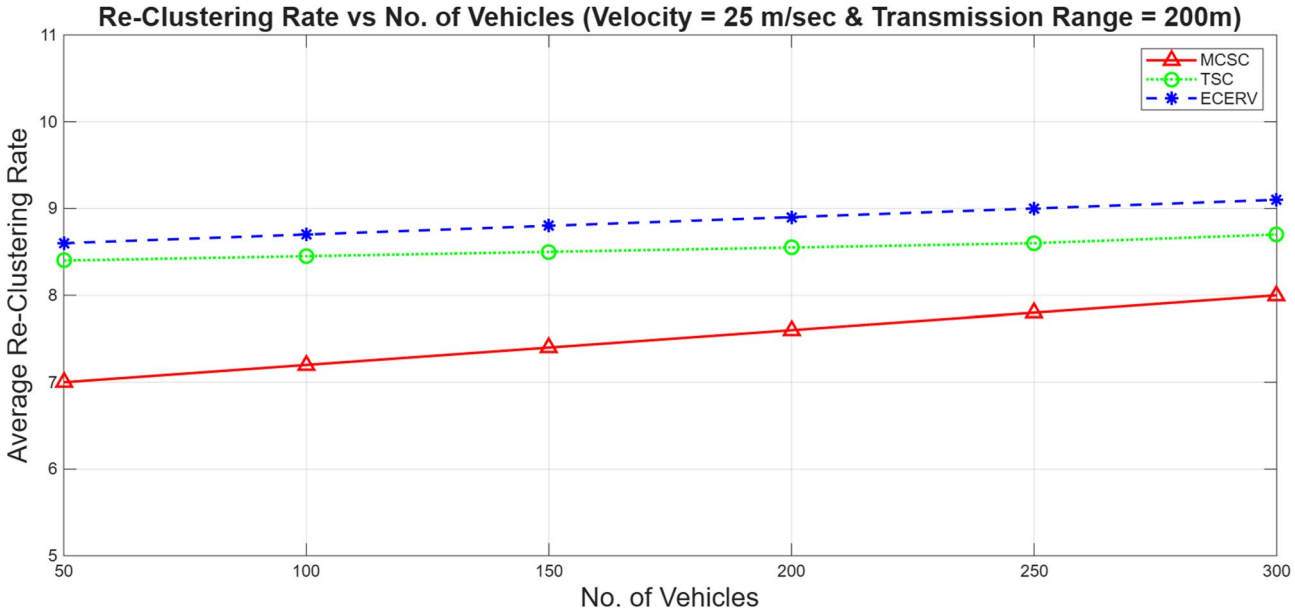


Fig. 12. Impact of number of vehicles on re-clustering rate.

technique is less affected by high mobility of vehicles as compared to ECERV[1] and TSC[13] mainly attributed to PCL based CH selection which favors more stable thereby performing well even in high mobility scenario.

Figure 5 shows performance of clustering techniques at different values of velocity in terms of CM duration. It is observed that the CM duration follows the same trend as CH duration because of high dynamic nature of vehicles it is challenging for CM to remain intact with CH in the form of clusters. By virtue of substitute CH, in case of temporary unavailability of primary CH, the proposed MCSC outperforms ECERV[1] as the cluster members would need to discover and join another cluster. In terms of CM duration, TSC[13] closely follows MCSC because both techniques introduced SCH as backup.

Figure 6 shows impact of velocity of vehicles on re-clustering rate. It is observed from the results that the re-clustering rate increases with the increase in velocity. The reason is that the increase in velocity accelerates the change of network topology. The performance of ECERV[1] is worst in terms of re-clustering rate as it makes clustering decisions on instantaneous network and mobility parameters which triggers frequent re-configurations with increase in speed. Both MCSC and TSC[13] make use of SCH thereby avoiding frequent re-clustering in high mobility. The relatively better performance of MCSC than TSC[13] can be attributed to its efficient CH selection based on CCF.

II. Impact of transmission range of vehicles.

This subsection describes how MCSC reacts at different transmission range of vehicles. In this experiment, the transmission range was varied from 200 m to 500 m.The velocity of vehicle was chosen as 25 m/s where the number of vehicles was kept at 200. The performance of MCSC against other techniques, with regard to CH and CM duration at different transmission range, has been shown in Figs. 7 and 8 respectively. From the results, it is observed that both CH and CM duration have large values at high-transmission range. This is because high transmission range allows the vehicles to connect with more neighboring vehicles. MCSC performs better than ECERV[1] and TSC[13] in this set of experiments because in our technique CLS is introduced to filter neighboring vehicle and form cluster with stable neighbors, which is directly proportional to transmission range.

Figure 9 represents re-clustering rate of MCSC compared to other techniques at different transmission range. Results show that re-clustering rate slightly decreases with the increase in transmission range because higher transmission range offers more connectivity among vehicles and mitigates the change of CH. MCSC outperforms ECERV[1] and TSC[13] due to its stability-aware multi-criterion CH selection.

III. Impact of number of vehicles.

This subsection evaluates the performance of MCSC when exposed to different network sizes. Figure 10 shows the performance of MCSC in terms of CH duration ratio where the total numbers of vehicles are varied from 50 to 300. In this experiment, the velocity of vehicle was kept at 25 m/s and the transmission range at 200 m. Results show the impact of number of vehicles on CH duration in which we observed that the increase in vehicles density

leads to higher contention and frequent topology changing, thereby reducing CH lifetime. Unlike existing techniques where cluster merging is performed based on predefined threshold values, MCSC merges clusters based on the CCF, thereby neighboring clusters are merged under the leadership of a more eligible cluster head.

Figure 11 represents the impact of varying number of vehicles on CM duration. It is observed that CM duration decreases with the increase in density of vehicles. The reason is that increase in number of vehicles in vicinity also increases qualified vehicles to become CH and afterwards CM disassociates with previous CH and re-associates with new CH, which eventually decrease CM duration. Although there is slight decrease but the duration in which vehicle is not associated with any CH is not significant. In MCSC, clusters merging is performed when more eligible vehicles enter into the proximity of existing clusters and therefore CM duration is less affected by the density of vehicle, resulting in improved performance of MCSC as compared to TSC[13] and ECERV[1].

Figure 12 illustrates the impact of different network sizes on re-clustering rate. Results show that with the increase in number of vehicles, re-clustering rate also increases in all three techniques. The reason is that increase in number of vehicles in vicinity also increases more eligible vehicle to compete for CH role which cause frequent change in CH role. MCSC has performed better than other techniques because of CCF based efficient criteria which doesn't allow vehicle with lower value to become CH and ultimately prevent frequent re-clustering.

Limitations and future work

This study focused on stable clustering for bidirectional road segments without intersections, assuming relatively smooth vehicular mobility and homogeneous communication conditions. Complex urban factors such as intersections, traffic signals, multi-lane turning behavior, and heterogeneous traffic density are not explicitly modeled. In addition, the scheme relies on fixed and uniformly weighted clustering metrics, which may not fully capture rapidly changing network dynamics in dense or highly irregular environments.

Part of future work we plan to extend the proposed approach to urban intersection scenarios by incorporating traffic-light awareness and turn-based mobility models. Further research may also consider heterogeneous vehicles, variable transmission ranges, and realistic channel impairments, as well as large-scale validation using real-world vehicular mobility traces. Another future direction would be to integrate lightweight authentication and conditional privacy-preserving mechanisms (e.g., pseudonym-based or certificate-free schemes) at the CH/SCH level, enabling secure message validation while maintaining user privacy without compromising clustering stability.

Conclusions

This paper proposed a Multi-Criteria-based Stable Clustering (MCSC) technique to enhance clustering stability in VANETs. The key contribution is a stability-aware clustering framework that integrates Connectivity Lifespan (CLS) and a Composite Clustering Function (CCF) incorporating a novel Past Cluster-Head Lifetime (PCL) metric for robust cluster-head selection. The inclusion of a Substitute Cluster Head (SCH) and an adaptive cluster merging mechanism further strengthens cluster continuity under high mobility.

Extensive simulations under varying vehicle speeds, transmission ranges, and network densities quantitatively demonstrate the effectiveness of MCSC. Compared to existing approaches, MCSC achieves approximately 16% and 14.5% improvements in average CH and CM lifetimes, respectively, and reduces the re-clustering rate by about 13%. Supported by simulation results, the proposed technique significantly improves clusters stability in highly dynamic vehicular environments.

Data availability

All data generated or analysed during this study are included in this published article.

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Author contributions
The core research idea was developed and formulated by Dr. A.W.K. He contributed in not only identifying the research gaps but also developed strategies for addressing the research questions. Overall project administration was also carried out by him throughout the work. Dr. J.I.B. performed all the simulations of the proposed technique and contributed in benchmarking with existing literature. Dr. S.K. conducted the validation as well as the comparative analysis while Mr. Z.H.B.A. has reviewed and edited the manuscript. All the authors have read and approved the final version of the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
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